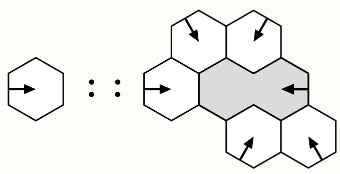
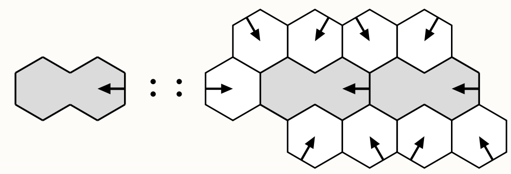
Mountain and Valley

a hat tiling variant

Mountain (H)



Valley (D)



Address Φ

$$
\begin{aligned}
D_i &\mapsto D_iK_i\\
K_i &\mapsto D_iK_iH_i\\
H_i &\mapsto D_iK_iH_i\\
a_i &\mapsto D_{i-2}a_iH_i\\
b_i &\mapsto D_{i-1}b_iH_i\\
g_i &\mapsto g_iH_i\\
h_i &\mapsto h_iH_i
\end{aligned}
$$

indices are cyclic modulo 6.

Materia

$$
H_i@X \longmapsto \begin{cases} D_i@\big(\Phi(X)+\langle 0\rangle\big), \\ H_j@\big(\Phi(X)+\big\langle D_iK_i,\ D_ia_{i+2},\ D_ib_{i+1}, \\ \quad g_{i-1},\ h_{i-2}\big\rangle\big); \end{cases}
$$

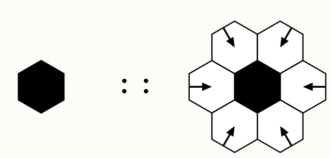
$$
D_i@X \longmapsto \begin{cases} D_i@\big(\Phi(X)+\langle 0,\ D_iK_i\rangle\big), \\ H_j@\big(\Phi(X)+\big\langle D_iK_iD_iK_i,\ D_ia_{i+2}, \\ \quad D_ib_{i+1},\ D_iK_iD_ia_{i+2},\ D_iK_iD_ib_{i+1}, \\ \quad g_{i-1},\ h_{i-2},\ D_iK_ig_{i-1},\ D_iK_ih_{i-2}\big\rangle\big). \end{cases}
$$

j follows the final step direction.

Axioms

name	specification
Single dimer	$D_i@0$
Single hexagon	$H_i@0$
Fixed-point seed	$D \mid H$
Branch C_3	$\langle H_4@e,\ H_0@D_1K_3,\ H_2@D_2K_4\rangle$
Leaf C_3	$\langle H_3@e,\ H_5@D_0K_2K_3,\ H_1@D_1K_3K_4\rangle$
False center	$C@0$

False center



$$C@0 \longmapsto C@0 + H_i@\langle g_i : i = 0, \ldots, 5\rangle.$$

Two inflations of the $D \mid H$ axiom